# Sakshi Nair

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## Summary


Data Scientist bridging geospatial/remote-sensing analysis and causal inference – from satellite-imagery classification and spatial pipelines to A/B testing and forecasting models – with production ML/data infrastructure and automated reporting on top. Builds statistically rigorous, deployed solutions that turn ambiguous business questions into measurable outcomes for non-technical stakeholders.


## Education

**Indiana University Bloomington – M.S. Data Science, GPA 3.9/4.0** **Aug 2024 – May 2026**
Coursework: Applied Machine Learning, Software Engineering, Data Visualization, Advanced Databases, Applied Algorithms.

**Symbiosis International University, Maharashtra, India – M.S. Applied Statistics** **May 2024**
Coursework: Statistical Machine Learning, Regression Analysis, Data Mining, Time Series Analysis, Optimization Techniques, Stochastic Processes.

**Osmania University, Telangana, India – B.S. Mathematics, Statistics, Computer Science** **Jun 2022**

## Technical Skills

**Languages:** Python, SQL, R, JavaScript, PL/pgSQL
**Machine Learning:** Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
**AI and LLMs:** LangGraph, LangChain, OpenAI API, ChromaDB, RAG, Pydantic, Streamlit
**Visualization and BI:** Tableau, Power BI, Looker, Looker Studio, Amazon QuickSight
**Data and Infrastructure:** PostgreSQL, Snowflake, dbt, Airflow, Docker, ArcGIS, Google Earth Engine, GeoPandas, Git, GitHub Actions
**Web and Cloud:** React, Leaflet, Node.js, Express, AWS (Athena, Lambda, Glue, Redshift, SageMaker, S3), Excel

## Experience

**Indiana University – O'Neill School – Research Assistant** **Jan 2026 – May 2026**
*Bloomington, IN*

- Built a global platform mapping 3,000+ floating solar installations across 80+ countries by fusing Sentinel-2 and Landsat-8 imagery through Google Earth Engine and Python/GeoPandas pipelines processing 500+ GB of satellite data; exact-overlap matching was dropping valid installations against messy real-world water-body boundaries, so switched to a buffered spatial join to correctly reconcile footprints.
- Shipped an interactive React/Leaflet front end on a Node/Express REST API backend (99% uptime), replacing manual custom data pulls with a self-serve platform non-technical researchers could query directly for on-demand metrics.

**Indiana University – Kelley School – Data Analyst** **Apr 2025 – May 2026**
*Bloomington, IN*

- Designed and analyzed pricing A/B tests (two-sample t-tests) measuring price sensitivity; presented results to non-technical stakeholders who approved the winning variant, cutting costs by $10K/month.
- Reduced false-positive alert noise while still catching genuine data issues by tuning anomaly detection thresholds on ARIMA/regression forecasting models (Python, AWS Lambda, Glue) monitoring 5,000+ daily Redshift transactions.

**Suhora Technologies – Data Scientist Intern** **Jan 2024 – Jul 2024**
*Delhi, India*

- Built-up and barren land classes had overlapping spectral signatures that were misclassifying pixels in initial ISODATA/Maximum Likelihood Classification attempts on 1985 and 2014 Hyderabad Landsat imagery; fixed by iterating on training-site selection in ArcGIS, improving accuracy to 85% (Kappa = 0.744) from an 80% baseline (Kappa = 0.719).
- Delivered a change-detection analysis quantifying 29 years of urban sprawl: 86.35% built-up increase, 24.25% agricultural decline, and 53.41% water-body reduction.

## Projects

**Northwind Retail Analytics** **SQL, PostgreSQL, Power BI, Looker, Tableau** | GitHub

- Designed a star-schema data warehouse (1 fact + 5 dimension tables, 77K+ transaction records across 45,000 orders) and rebuilt an identical 4-view analytics suite across Power BI, Looker, and Tableau, authoring 15+ tool-specific measures (DAX, LookML, calculated fields).
- Validating at the monthly grain against SQL ground-truth queries, not just lifetime totals, surfaced a case where Looker's `dimension_group` defaulted to UTC and shifted month-end orders into the next month while Power BI and Tableau's naive date bucketing didn't; fixed by aligning LookML's model timezone to the source data, achieving exact parity across all 12 published dashboards.

**CommerceLens – E-Commerce SQL Data Warehouse** **PostgreSQL, PL/pgSQL, dbt, Docker, Metabase** | GitHub

- Designed a star-schema warehouse in PostgreSQL and built both a hand-written SQL ETL and an equivalent dbt project (20 models, 50/50 tests passing) loading 25,000 orders and 74,859 line items; since a bug in either pipeline surfaces as a silent mismatch, wrote a verification harness diffing row counts and checksums between both outputs on every run.
- Indexing `fact_returns` for a broad refund-by-state-and-reason report barely moved the needle since Postgres correctly preferred a sequential scan across most of both tables; the real payoff was a high-selectivity single-order lookup (˜1:5000), cut from 2.16ms to 0.31ms (Seq Scan → Bitmap Index Scan) via an EXPLAIN ANALYZE case study.

**MarketPulse AI – Multi-Agent Competitive Intelligence System** **Python, LangGraph, OpenAI, Slack API, SQLite** | GitHub

- Orchestrated a multi-agent LangGraph graph, weekday-scheduled via GitHub Actions: a planner node decomposes a company watchlist into research queries, fans them to parallel per-competitor agents, then a verification agent marks a claim verified only when two sources agree, before pausing for Slack approval; averaged $0.0048/run at a 60% first-pass success rate across test runs.
- LangGraph re-executes a paused node's function on resume, so a Slack alert inside the approval node fired twice per run; caught via log timestamps, fixed by moving the side effect to the once-only synthesis step, locked in with a test asserting delivery only after approval.